

Trajectory_Analysis_Perturbation_Cleaned

June 1, 2026

1 Introduction to Single-Cell Perturbation Analysis

Single-cell perturbation analysis is a framework for identifying how a biological treatment, stimulus, or genetic perturbation affects cellular states at the single-cell level. Rather than comparing bulk averages between treated and control samples, this approach leverages single-cell resolution to quantify subtle and cell-type-specific responses to a perturbation.

Perturbation analysis in single-cell data often revolves around the idea of separability: if a perturbation, could be chemical, biological or genetic (CRISPRi) has a strong effect on a given cell population, it should be easy to distinguish treated from untreated cells based on their transcriptomes. Conversely, if the perturbation barely affects a cell type, classification between treated and untreated will be difficult.

In our case, we have two single-cell RNA-seq datasets: one representing cells exposed to interferon beta (IFN- β) after 6 hours and another representing unexposed control cells. The goal is to determine which cell types are most responsive to interferon signaling and how strongly their transcriptional profiles shift in response.

1.0.1 Installations

```
[ ]: !pip install scanpy
      !pip install anndata
      !pip3 install igraph
      !pip install celltypist
      !pip install decoupler
      !pip install fa2-modified
      !pip install louvain
      !pip install perpty
```

Collecting scanpy

Downloading scanpy-1.11.5-py3-none-any.whl.metadata (9.3 kB)

Collecting anndata>=0.8 (from scanpy)

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Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.15.1)

Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.5.2)

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 Requirement already satisfied: numpy>=1.24.1 in /usr/local/lib/python3.12/dist-
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 (from scanpy) (4.67.1)
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 /usr/local/lib/python3.12/dist-packages (from scanpy) (4.15.0)
 Requirement already satisfied: umap-learn>=0.5.6 in
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 Collecting array-api-compat>=1.7.1 (from anndata>=0.8->scanpy)
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 Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-
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 Requirement already satisfied: fonttools>=4.22.0 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy)
 (4.60.1)
 Requirement already satisfied: kiwisolver>=1.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (2.9.0.post0)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba!=0.62.0rc1,>=0.57.1->scanpy) (0.43.0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5.3->scanpy) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5.3->scanpy) (2025.2)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.1.3->scanpy) (3.6.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.7.5->scanpy) (1.17.0)

Collecting donfig>=0.8 (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)

Downloading donfig-0.8.1.post1-py3-none-any.whl.metadata (5.0 kB)

Collecting numcodecs>=0.14 (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)

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Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy) (6.0.3)

Collecting crc32c>=2.7 (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)

Downloading crc32c-2.8-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (7.8 kB)

Downloading scanpy-1.11.5-py3-none-any.whl (2.1 MB)

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Downloading legacy_api_wrap-1.4.1-py3-none-any.whl (10.0 kB)

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Downloading array_api_compat-1.12.0-py3-none-any.whl (58 kB)

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Downloading zarr-3.1.3-py3-none-any.whl (276 kB)

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Downloading donfig-0.8.1.post1-py3-none-any.whl (21 kB)

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numcodecs-0.16.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (8.8 MB)

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Downloading crc32c-2.8-cp312-cp312-

manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (80 kB)

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Installing collected packages: session-info2, numcodecs, legacy-api-wrap, donfig, crc32c, array-api-compat, zarr, anndata, scanpy

Successfully installed anndata-0.12.4 array-api-compat-1.12.0 crc32c-2.8 donfig-0.8.1.post1 legacy-api-wrap-1.4.1 numcodecs-0.16.3 scanpy-1.11.5 session-info2-0.2.3 zarr-3.1.3

Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-packages (0.12.4)

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Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-packages (from anndata) (3.15.1)

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Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from anndata) (8.4.0)

Requirement already satisfied: numpy>=1.26 in /usr/local/lib/python3.12/dist-packages (from anndata) (2.0.2)

Requirement already satisfied: packaging>=24.2 in /usr/local/lib/python3.12/dist-packages (from anndata) (25.0)

Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from anndata) (2.2.2)

Requirement already satisfied: scipy>=1.12 in /usr/local/lib/python3.12/dist-packages (from anndata) (1.16.2)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata) (3.1.3)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2025.2)

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Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata) (6.0.3)

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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas!=2.1.2,>=2.1.0->anndata) (1.17.0)

Collecting igraph

Downloading igraph-1.0.0-cp39-abi3-manylinux_2_28_x86_64.whl.metadata (4.4 kB)

Collecting texttable>=1.6.2 (from igraph)

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Downloading texttable-1.7.0-py2.py3-none-any.whl (10 kB)

Installing collected packages: texttable, igraph

Successfully installed igraph-1.0.0 texttable-1.7.0

Collecting celltypist

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Requirement already satisfied: pandas>=1.0.5 in /usr/local/lib/python3.12/dist-packages (from celltypist) (2.2.2)

Requirement already satisfied: scikit-learn>=0.24.1 in /usr/local/lib/python3.12/dist-packages (from celltypist) (1.6.1)

Requirement already satisfied: openpyxl>=3.0.4 in /usr/local/lib/python3.12/dist-packages (from celltypist) (3.1.5)

Requirement already satisfied: click>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from celltypist) (8.3.0)

Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from celltypist) (2.32.4)

Requirement already satisfied: scanpy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from celltypist) (1.11.5)

Collecting leidenalg>=0.9.0 (from celltypist)

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Collecting igraph<0.12,>=0.10.0 (from leidenalg>=0.9.0->celltypist)

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Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl>=3.0.4->celltypist) (2.0.0)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celltypist) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celltypist) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celttypist) (2025.2)

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celttypist) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celttypist) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celttypist) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celttypist) (2025.10.5)

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Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (3.15.1)

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Requirement already satisfied: networkx>=2.7.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (3.5)

Requirement already satisfied: numba!=0.62.0rc1,>=0.57.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.60.0)

Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (25.0)

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Requirement already satisfied: pynndescent>=0.5.13 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.5.13)

Requirement already satisfied: scipy>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (1.16.2)

Requirement already satisfied: seaborn>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.13.2)

Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.2.3)

Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.14.5)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (4.67.1)

Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (4.15.0)

Requirement already satisfied: umap-learn>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.5.9.post2)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.1->celltypist) (3.6.0)

Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy>=1.7.0->celltypist) (1.12.0)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy>=1.7.0->celltypist) (3.1.3)

Requirement already satisfied: texttable>=1.6.2 in /usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->leidenalg>=0.9.0->celltypist) (1.7.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (3.2.5)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba!=0.62.0rc1,>=0.57.1->scanpy>=1.7.0->celltypist) (0.43.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.0.5->celltypist) (1.17.0)

Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.8.1.post1)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.16.3)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages

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(from
donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist)
(6.0.3)
Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-
packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8-
>scanpy>=1.7.0->celltypist) (2.8)
Downloading celltypist-1.7.1-py3-none-any.whl (7.3 MB)
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41.9 MB/s eta 0:00:00
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leidenalg-0.10.2-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.0
MB)
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46.0 MB/s eta 0:00:00
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igraph-0.11.9-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.4 MB)
4.4/4.4 MB
64.3 MB/s eta 0:00:00
Installing collected packages: igraph, leidenalg, celltypist
  Attempting uninstall: igraph
    Found existing installation: igraph 1.0.0
    Uninstalling igraph-1.0.0:
      Successfully uninstalled igraph-1.0.0
Successfully installed celltypist-1.7.1 igraph-0.11.9 leidenalg-0.10.2
Collecting decoupler
  Downloading decoupler-2.1.1-py3-none-any.whl.metadata (8.6 kB)
Collecting adjusttext (from decoupler)
  Downloading adjustText-1.3.0-py3-none-any.whl.metadata (3.1 kB)
Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-
packages (from decoupler) (0.12.4)
Collecting docrep (from decoupler)
  Downloading docrep-0.3.2.tar.gz (33 kB)
  Preparing metadata (setup.py) ... done
Collecting marsilea (from decoupler)
  Downloading marsilea-0.5.6-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packages
(from decoupler) (0.60.0)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
packages (from decoupler) (2.32.4)
Collecting scipy<1.16 (from decoupler)
  Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(61 kB)
62.0/62.0 kB
3.8 MB/s eta 0:00:00
Requirement already satisfied: session-info2 in
/usr/local/lib/python3.12/dist-packages (from decoupler) (0.2.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages

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(from decoupler) (4.67.1)
Requirement already satisfied: numpy<2.5,>=1.23.5 in
/usr/local/lib/python3.12/dist-packages (from scipy<1.16->decoupler) (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-
packages (from adjusttext->decoupler) (3.10.0)
Requirement already satisfied: array-api-compat>=1.7.1 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.12.0)
Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: legacy-api-wrap in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.4.1)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (8.4.0)
Requirement already satisfied: packaging>=24.2 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (25.0)
Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (2.2.2)
Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (3.1.3)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
(from docrep->decoupler) (1.17.0)
Collecting legendkit (from marsilea->decoupler)
  Downloading legendkit-0.3.6-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: platformdirs in /usr/local/lib/python3.12/dist-
packages (from marsilea->decoupler) (4.5.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-
packages (from marsilea->decoupler) (0.13.2)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from numba->decoupler) (0.43.0)
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/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->decoupler) (3.11)
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/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2025.10.5)
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/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(1.3.3)
Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(1.4.9)

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Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)

Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.8.1.post1)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.16.3)

Requirement already satisfied: typing-extensions>=4.9 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata->decoupler) (4.15.0)

Requirement already satisfied: pyarrow>=10.0.1 in /usr/local/lib/python3.12/dist-packages (from pandas[parquet]->marsilea->decoupler) (18.1.0)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (6.0.3)

Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (2.8)

Downloading decoupler-2.1.1-py3-none-any.whl (117 kB)
118.0/118.0 kB

6.5 MB/s eta 0:00:00

Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (37.3 MB)
37.3/37.3 MB

26.0 MB/s eta 0:00:00

Downloading adjustText-1.3.0-py3-none-any.whl (13 kB)

Downloading marsilea-0.5.6-py3-none-any.whl (84 kB)
85.0/85.0 kB

7.3 MB/s eta 0:00:00

Downloading legendkit-0.3.6-py3-none-any.whl (24 kB)

Building wheels for collected packages: docrep

Building wheel for docrep (setup.py) ... done

Created wheel for docrep: filename=docrep-0.3.2-py3-none-any.whl size=19876 sha256=a78df0360387c663a138276eedc87da1422a150bf0b0862479f70df94ba2f558

Stored in directory: /root/.cache/pip/wheels/d6/19/ee/0a6a1793d91c449563b49cca b57ce52da3e6fab7614836bd8c

Successfully built docrep

```

Installing collected packages: scipy, docrep, legendkit, adjusttext, marsilea,
decoupler
  Attempting uninstall: scipy
    Found existing installation: scipy 1.16.2
    Uninstalling scipy-1.16.2:
      Successfully uninstalled scipy-1.16.2
Successfully installed adjusttext-1.3.0 decoupler-2.1.1 docrep-0.3.2
legendkit-0.3.6 marsilea-0.5.6 scipy-1.15.3
Collecting fa2-modified
  Downloading fa2_modified-0.4-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (9.3
kB)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (1.15.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (4.67.1)
Downloading fa2_modified-0.4-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (620 kB)
      620.8/620.8 kB
8.0 MB/s eta 0:00:00
Installing collected packages: fa2-modified
Successfully installed fa2-modified-0.4
Collecting louvain
  Downloading louvain-0.8.2.tar.gz (4.2 MB)
      4.2/4.2 MB
26.3 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: igraph<0.12,>=0.10.0 in
/usr/local/lib/python3.12/dist-packages (from louvain) (0.11.9)
Requirement already satisfied: texttable>=1.6.2 in
/usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->louvain)
(1.7.0)
Building wheels for collected packages: louvain
  Building wheel for louvain (setup.py) ... done
  Created wheel for louvain: filename=louvain-0.8.2-cp312-cp312-linux_x86_64.whl
size=971849
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  Stored in directory: /root/.cache/pip/wheels/40/de/2b/bb7ed19d84727f9f299f20cd
34c42bba9c8bef7d83d2255c86
Successfully built louvain
Installing collected packages: louvain
Successfully installed louvain-0.8.2
Collecting perpty
  Downloading perpty-1.0.3-py3-none-any.whl.metadata (9.4 kB)
Requirement already satisfied: adjusttext in /usr/local/lib/python3.12/dist-
packages (from perpty) (1.3.0)

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Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from adjusttext->pertpy) (3.10.0)

Requirement already satisfied: setuptools>=60.0.0 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (75.2.0)

Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (25.0)

Requirement already satisfied: pandas>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (2.2.2)

Requirement already satisfied: xarray>=2023.7.0 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (2025.10.1)

Requirement already satisfied: h5netcdf>=1.0.2 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (1.7.3)

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Requirement already satisfied: xarray-einstats>=0.3 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (0.9.1)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from blitzgsea->pertpy) (4.67.1)

Requirement already satisfied: statsmodels in /usr/local/lib/python3.12/dist-packages (from blitzgsea->pertpy) (0.14.5)

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Requirement already satisfied: numba>=0.57 in /usr/local/lib/python3.12/dist-packages (from fast-array-utils[accel,sparse]->pertpy) (0.60.0)

Requirement already satisfied: jax>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.7.2)

Requirement already satisfied: msgpack in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (1.1.2)

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Requirement already satisfied: orbox-checkpoint in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.11.26)

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Requirement already satisfied: treescope>=0.1.7 in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.1.10)

Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->pertpy) (4.0.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->pertpy) (2.19.2)

Collecting makefun (from funsor->pertpy)

 Downloading makefun-1.16.0-py2.py3-none-any.whl.metadata (2.9 kB)

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Requirement already satisfied: anndata>=0.10.8 in
 /usr/local/lib/python3.12/dist-packages (from mudata->pertpy) (0.12.4)

Requirement already satisfied: jaxlib>=0.4.25 in /usr/local/lib/python3.12/dist-
 packages (from numpyro->pertpy) (0.7.2)

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 packages (from openpyxl->pertpy) (2.0.0)

Collecting jaxopt>=0.8 (from ott-jax->pertpy)
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Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: certifi>=2017.4.17 in
 /usr/local/lib/python3.12/dist-packages (from requests->pertpy) (2025.10.5)

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Requirement already satisfied: legacy-api-wrap>=1.4.1 in
 /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (1.4.1)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: umap-learn>=0.5.6 in
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 (from sparsecca->pertpy) (6.9.5)

Requirement already satisfied: array-api-compat>=1.7.1 in
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 (1.12.0)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
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 (3.1.3)

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Collecting equinox>=0.11.10 (from lineax>=0.0.7->ott-jax->pertpy)
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Collecting jaxtyping>=0.2.24 (from lineax>=0.0.7->ott-jax->pertpy)
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 Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy) (1.3.3)
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 Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy) (2.9.0.post0)
 Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba>=0.57->fast-array-utils[accel,sparse]->pertpy) (0.43.0)
 Requirement already satisfied: absl-py>=0.7.1 in /usr/local/lib/python3.12/dist-packages (from optax->flax->pertpy) (1.4.0)
 Requirement already satisfied: chex>=0.1.87 in /usr/local/lib/python3.12/dist-packages (from optax->flax->pertpy) (0.1.90)
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 Requirement already satisfied: etils[epath,epy] in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (1.13.0)
 Requirement already satisfied: nest_asyncio in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (1.6.0)
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Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (5.9.5)

Requirement already satisfied: ply in /usr/local/lib/python3.12/dist-packages (from pyomo->sparseccca->pertpy) (3.11)

Requirement already satisfied: toolz>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from chex>=0.1.87->optax->flax->pertpy) (0.12.1)

Collecting wadler-lindig>=0.1.0 (from equinox>=0.11.10->lineax>=0.0.7->ott-jax->pertpy)

Downloading wadler_lindig-0.1.7-py3-none-any.whl.metadata (17 kB)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->adjusttext->pertpy) (1.17.0)

Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->mudata->pertpy) (0.8.1.post1)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->mudata->pertpy) (0.16.3)

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Requirement already satisfied: importlib_resources in /usr/local/lib/python3.12/dist-packages (from etils[epath,epy]->orbax-checkpoint->flax->pertpy) (6.5.2)

Requirement already satisfied: zipp in /usr/local/lib/python3.12/dist-packages (from etils[epath,epy]->orbax-checkpoint->flax->pertpy) (3.23.0)

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Downloading pertpy-1.0.3-py3-none-any.whl (216 kB)

216.9/216.9 kB

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Downloading

scikit_misc-0.5.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl


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Building wheels for collected packages: blitzgsea
  Building wheel for blitzgsea (setup.py) ... done
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  Stored in directory: /root/.cache/pip/wheels/e3/00/84/15131449d6f397371baa9561
  4491906dbd4ca2a932194b9494
Successfully built blitzgsea
Installing collected packages: makefun, lamin-utils, wadler-lindig, scikit-misc,
pubchempy, funsor, fast-array-utils, jaxtyping, sparsecca, numpyro, jaxopt,
equinox, blitzgsea, mudata, lineax, ott-jax, perptpy
Successfully installed blitzgsea-1.3.54 equinox-0.13.2 fast-array-utils-1.3
funsor-0.4.5 jaxopt-0.8.5 jaxtyping-0.3.3 lamin-utils-0.15.0 lineax-0.0.8
makefun-1.16.0 mudata-0.3.2 numpyro-0.19.0 ott-jax-0.5.2 perptpy-1.0.3
pubchempy-1.0.5 scikit-misc-0.5.1 sparsecca-0.3.1 wadler-lindig-0.1.7

```

```
[ ]: !pip install harmonypy
```

```

Collecting harmonypy
  Downloading harmonypy-0.0.10-py3-none-any.whl.metadata (3.9 kB)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from harmonypy) (2.0.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
(from harmonypy) (2.2.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-
packages (from harmonypy) (1.6.1)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from harmonypy) (1.15.3)
Requirement already satisfied: python-dateutil>=2.8.2 in

```

```

/usr/local/lib/python3.12/dist-packages (from pandas->harmonypy) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas->harmonypy) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas->harmonypy) (2025.2)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-
packages (from scikit-learn->harmonypy) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn->harmonypy) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.8.2->pandas->harmonypy) (1.17.0)
Downloading harmonypy-0.0.10-py3-none-any.whl (20 kB)
Installing collected packages: harmonypy
Successfully installed harmonypy-0.0.10

```

1.0.2 Import Libraries

```

[ ]: #Import core single cell datasets

import scanpy as sc
import anndata as ad
import numpy as np
import pertpy as pt
import pandas as pd

```

1.0.3 Download the cells without IFN stimulation

```

[ ]: !wget https://github.com/josoga2/sc/raw/refs/heads/main/GSE96583_RAW/
↳GSM2560249_2.2.mtx.gz
!wget https://github.com/josoga2/sc/raw/refs/heads/main/GSE96583_RAW/
↳GSM2560249_barcodes.tsv.gz
!wget https://github.com/josoga2/sc/raw/refs/heads/main/GSE96583_RAW/
↳GSE96583_batch2.genes.tsv.gz

```

```

--2025-10-29 04:32:30-- https://github.com/josoga2/sc/raw/refs/heads/main/GSE96
583_RAW/GSM2560249_2.2.mtx.gz
Resolving github.com (github.com)... 20.27.177.113
Connecting to github.com (github.com)|20.27.177.113|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/josoga2/sc/refs/heads/main/GSE96583_
RAW/GSM2560249_2.2.mtx.gz [following]
--2025-10-29 04:32:30-- https://raw.githubusercontent.com/josoga2/sc/refs/heads
/main/GSE96583_RAW/GSM2560249_2.2.mtx.gz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.110.133, 185.199.109.133, 185.199.108.133, ...

```

```

Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.110.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 29050932 (28M) [application/octet-stream]
Saving to: 'GSM2560249_2.2.mtx.gz'

GSM2560249_2.2.mtx. 100%[=====>] 27.71M 66.2MB/s in 0.4s

2025-10-29 04:32:33 (66.2 MB/s) - 'GSM2560249_2.2.mtx.gz' saved
[29050932/29050932]

--2025-10-29 04:32:33-- https://github.com/josoga2/sc/raw/refs/heads/main/GSE96
583_RAW/GSM2560249_barcodes.tsv.gz
Resolving github.com (github.com)... 20.27.177.113
Connecting to github.com (github.com)|20.27.177.113|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/josoga2/sc/refs/heads/main/GSE96583_
RAW/GSM2560249_barcodes.tsv.gz [following]
--2025-10-29 04:32:34-- https://raw.githubusercontent.com/josoga2/sc/refs/heads
/main/GSE96583_RAW/GSM2560249_barcodes.tsv.gz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 52366 (51K) [application/octet-stream]
Saving to: 'GSM2560249_barcodes.tsv.gz'

GSM2560249_barcodes 100%[=====>] 51.14K --.-KB/s in 0.04s

2025-10-29 04:32:34 (1.31 MB/s) - 'GSM2560249_barcodes.tsv.gz' saved
[52366/52366]

--2025-10-29 04:32:34-- https://github.com/josoga2/sc/raw/refs/heads/main/GSE96
583_RAW/GSE96583_batch2.genes.tsv.gz
Resolving github.com (github.com)... 20.27.177.113
Connecting to github.com (github.com)|20.27.177.113|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/josoga2/sc/refs/heads/main/GSE96583_
RAW/GSE96583_batch2.genes.tsv.gz [following]
--2025-10-29 04:32:34-- https://raw.githubusercontent.com/josoga2/sc/refs/heads
/main/GSE96583_RAW/GSE96583_batch2.genes.tsv.gz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 277054 (271K) [application/octet-stream]

```

Saving to: 'GSE96583_batch2.genes.tsv.gz'

GSE96583_batch2.gen 100%[=====>] 270.56K --.-KB/s in 0.09s

2025-10-29 04:32:35 (2.88 MB/s) - 'GSE96583_batch2.genes.tsv.gz' saved
[277054/277054]

```
[ ]: !mkdir -p GSM2560245_Before
!mv GSM2560249_2.2.mtx.gz /content/GSM2560245_Before
!mv GSM2560249_barcodes.tsv.gz /content/GSM2560245_Before
!mv GSE96583_batch2.genes.tsv.gz /content/GSM2560245_Before

#rename files to matrix, barcode and genes
!mv /content/GSM2560245_Before/GSM2560249_2.2.mtx.gz /content/GSM2560245_Before/
↪matrix.mtx.gz
!mv /content/GSM2560245_Before/GSM2560249_barcodes.tsv.gz /content/
↪GSM2560245_Before/barcodes.tsv.gz
!mv /content/GSM2560245_Before/GSE96583_batch2.genes.tsv.gz /content/
↪GSM2560245_Before/genes.tsv.gz
```

1.0.4 Download the cells with IFN stimulation after 6 hrs

```
[ ]: !wget https://github.com/josoga2/sc/raw/refs/heads/main/GSE96583_RAW/
↪GSM2560248_2.1.mtx.gz
!wget https://github.com/josoga2/sc/raw/refs/heads/main/GSE96583_RAW/
↪GSM2560248_barcodes.tsv.gz
!wget https://github.com/josoga2/sc/raw/refs/heads/main/GSE96583_RAW/
↪GSE96583_batch2.genes.tsv.gz
```

```
--2025-10-29 04:32:36-- https://github.com/josoga2/sc/raw/refs/heads/main/GSE96583_RAW/GSM2560248_2.1.mtx.gz
Resolving github.com (github.com)... 20.27.177.113
Connecting to github.com (github.com)|20.27.177.113|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/josoga2/sc/refs/heads/main/GSE96583_RAW/GSM2560248_2.1.mtx.gz [following]
--2025-10-29 04:32:36-- https://raw.githubusercontent.com/josoga2/sc/refs/heads/main/GSE96583_RAW/GSM2560248_2.1.mtx.gz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 28545918 (27M) [application/octet-stream]
Saving to: 'GSM2560248_2.1.mtx.gz'
```

GSM2560248_2.1.mtx. 100%[=====>] 27.22M 65.5MB/s in 0.4s

2025-10-29 04:32:38 (65.5 MB/s) - 'GSM2560248_2.1.mtx.gz' saved
[28545918/28545918]

--2025-10-29 04:32:38-- https://github.com/josoga2/sc/raw/refs/heads/main/GSE96583_RAW/GSM2560248_barcodes.tsv.gz
Resolving github.com (github.com)... 20.27.177.113
Connecting to github.com (github.com)|20.27.177.113|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/josoga2/sc/refs/heads/main/GSE96583_RAW/GSM2560248_barcodes.tsv.gz [following]
--2025-10-29 04:32:39-- https://raw.githubusercontent.com/josoga2/sc/refs/heads/main/GSE96583_RAW/GSM2560248_barcodes.tsv.gz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 52904 (52K) [application/octet-stream]
Saving to: 'GSM2560248_barcodes.tsv.gz'

GSM2560248_barcodes 100%[=====>] 51.66K --.-KB/s in 0.04s

2025-10-29 04:32:39 (1.25 MB/s) - 'GSM2560248_barcodes.tsv.gz' saved
[52904/52904]

--2025-10-29 04:32:39-- https://github.com/josoga2/sc/raw/refs/heads/main/GSE96583_RAW/GSE96583_batch2.genes.tsv.gz
Resolving github.com (github.com)... 20.27.177.113
Connecting to github.com (github.com)|20.27.177.113|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/josoga2/sc/refs/heads/main/GSE96583_RAW/GSE96583_batch2.genes.tsv.gz [following]
--2025-10-29 04:32:39-- https://raw.githubusercontent.com/josoga2/sc/refs/heads/main/GSE96583_RAW/GSE96583_batch2.genes.tsv.gz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 277054 (271K) [application/octet-stream]
Saving to: 'GSE96583_batch2.genes.tsv.gz'

GSE96583_batch2.gen 100%[=====>] 270.56K --.-KB/s in 0.09s

2025-10-29 04:32:40 (2.86 MB/s) - 'GSE96583_batch2.genes.tsv.gz' saved

[277054/277054]

```
[ ]: !mkdir -p GSM2560245_After
!mv GSM2560248_2.1.mtx.gz /content/GSM2560245_After
!mv GSM2560248_barcodees.tsv.gz /content/GSM2560245_After
!mv GSE96583_batch2.genes.tsv.gz /content/GSM2560245_After

#rename the files accordingly
!mv /content/GSM2560245_After/GSM2560248_2.1.mtx.gz /content/GSM2560245_After/
↪matrix.mtx.gz
!mv /content/GSM2560245_After/GSM2560248_barcodees.tsv.gz /content/
↪GSM2560245_After/barcodees.tsv.gz
!mv /content/GSM2560245_After/GSE96583_batch2.genes.tsv.gz /content/
↪GSM2560245_After/genes.tsv.gz
```

```
[ ]: #write a function for assembling the dataset

def load_one_dataset(path, condition_label):
    # Read matrix
    adata = sc.read_mtx(f"{path}/matrix.mtx.gz").T

    # Read barcodees
    adata.obs_names = pd.read_csv(f"{path}/barcodees.tsv.gz", header=None,
↪sep="\t")[0]

    # Read and clean genes
    genes = pd.read_csv(f"{path}/genes.tsv.gz", header=None, sep = '\t')[1]
    #print(genes.head())
    genes = genes.astype(str).str.replace(r'^\d+', '', regex=True)
    adata.var_names = genes.var_names = genes[:adata.shape[1]]

    #make unique
    adata.var_names_make_unique()
    adata.obs_names_make_unique()

    # Add condition label
    adata.obs["condition"] = condition_label

    return adata
```

```
[ ]: adata_before = load_one_dataset("/content/GSM2560245_After", "before_IFN_beta")
adata_after = load_one_dataset("/content/GSM2560245_Before", "after_IFN_beta")
```

```
[ ]: ifn_adata = adata_before.concatenate(adata_after, batch_key="condition",
↪batch_categories=["before_IFN_beta",
↪"after_IFN_beta"])
```

/tmp/ipython-input-7954371.py:1: FutureWarning: Use anndata.concat instead of AnnData.concatenate, AnnData.concatenate is deprecated and will be removed in the future. See the tutorial for concat at:

<https://anndata.readthedocs.io/en/latest/concatenation.html>

```
ifn_adata = adata_before.concatenate(adata_after, batch_key="condition",
```

```
[ ]: ifn_adata.var.head()
```

```
[ ]: Empty DataFrame
```

```
Columns: []
```

```
Index: [MIR1302-10, FAM138A, OR4F5, RP11-34P13.7, RP11-34P13.8]
```

1.0.5 The Core Pipeline!

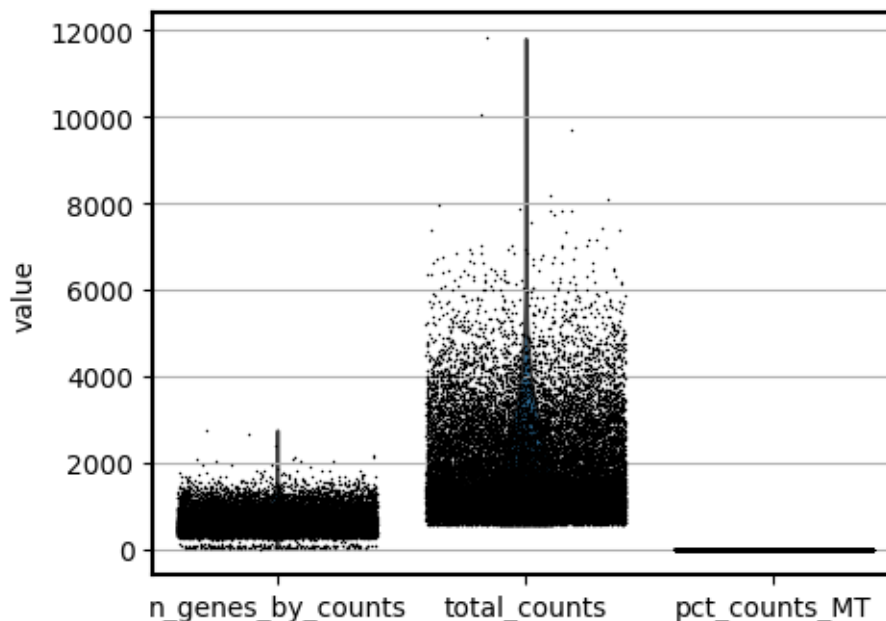
```
[ ]: ifn_adata.var['MT'] = ifn_adata.var_names.str.startswith("MT-")
ifn_adata.var['RIBO'] = ifn_adata.var_names.str.startswith("RPS", "RPL")
ifn_adata.var['HB'] = ifn_adata.var_names.str.startswith("^HB[^(P)]")
```

```
[ ]: sc.pp.calculate_qc_metrics(
    ifn_adata, qc_vars=["MT", 'RIBO', 'HB'], inplace=True, log1p=True
)
```

```
[ ]: import matplotlib.pyplot as plt

plt.rcParams["figure.figsize"] = (5,4) # Adjust figure size
plt.rcParams["axes.grid"] = True # Add grid to plots
plt.rcParams["axes.edgecolor"] = "black" # Set plot border color
plt.rcParams["axes.linewidth"] = 1.5 # Set plot border width
plt.rcParams["axes.facecolor"] = "white" # Set background color
plt.rcParams["axes.labelcolor"] = "black" # Set label color
plt.rcParams["xtick.color"] = "black" # Set x-axis tick color
plt.rcParams["ytick.color"] = "black" # Set y-axis tick color
plt.rcParams["text.color"] = "black" # Set text color
%matplotlib inline
```

```
[ ]: sc.pl.violin(
    ifn_adata,
    ["n_genes_by_counts", 'total_counts', 'pct_counts_MT'],
    jitter=0.4,
    multi_panel=False,
)
```



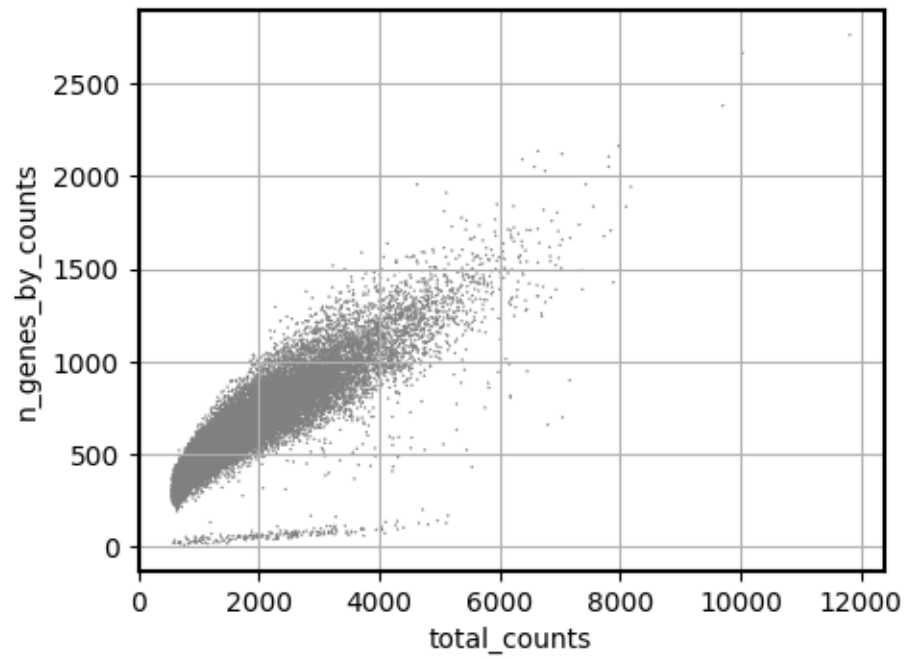
```
[ ]: ifn_adata.obs_keys()
```

```
/tmp/ipython-input-969156741.py:1: FutureWarning: Use obs (e.g. `k` in adata.obs`
or `str(adata.obs.columns.tolist())`) instead of AnnData.obs_keys,
AnnData.obs_keys is deprecated and will be removed in the future.
  ifn_adata.obs_keys()
```

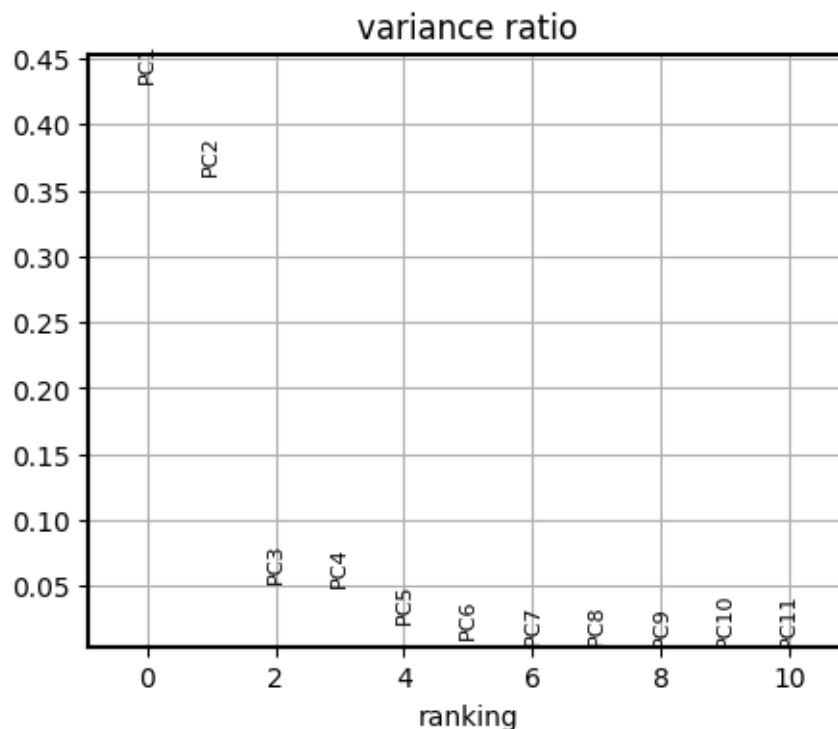
```
[ ]: ['condition',
      'n_genes_by_counts',
      'log1p_n_genes_by_counts',
      'total_counts',
      'log1p_total_counts',
      'pct_counts_in_top_50_genes',
      'pct_counts_in_top_100_genes',
      'pct_counts_in_top_200_genes',
      'pct_counts_in_top_500_genes',
      'total_counts_MT',
      'log1p_total_counts_MT',
      'pct_counts_MT',
      'total_counts_RIBO',
      'log1p_total_counts_RIBO',
      'pct_counts_RIBO',
      'total_counts_HB',
      'log1p_total_counts_HB',
      'pct_counts_HB']
```



```
[ ]: sc.pl.scatter(ifn_adata, "total_counts", "n_genes_by_counts")
```



```
[ ]: #Dim Reduction  
sc.tl.pca(ifn_adata)  
sc.pl.pca_variance_ratio(ifn_adata, n_pcs=10, log=False)
```



```
[ ]: #Let's correct for batch effect
import scanpy.external as sce
sce.pp.harmony_integrate(ifn_adata, key="condition")
```

2025-10-29 04:33:33,670 - harmony - INFO - Computing initial centroids with sklearn.KMeans...

INFO:harmony:Computing initial centroids with sklearn.KMeans...

2025-10-29 04:33:44,843 - harmony - INFO - sklearn.KMeans initialization complete.

INFO:harmony:sklearn.KMeans initialization complete.

2025-10-29 04:33:45,155 - harmony - INFO - Iteration 1 of 10

INFO:harmony:Iteration 1 of 10

2025-10-29 04:33:58,817 - harmony - INFO - Iteration 2 of 10

INFO:harmony:Iteration 2 of 10

2025-10-29 04:34:11,578 - harmony - INFO - Iteration 3 of 10

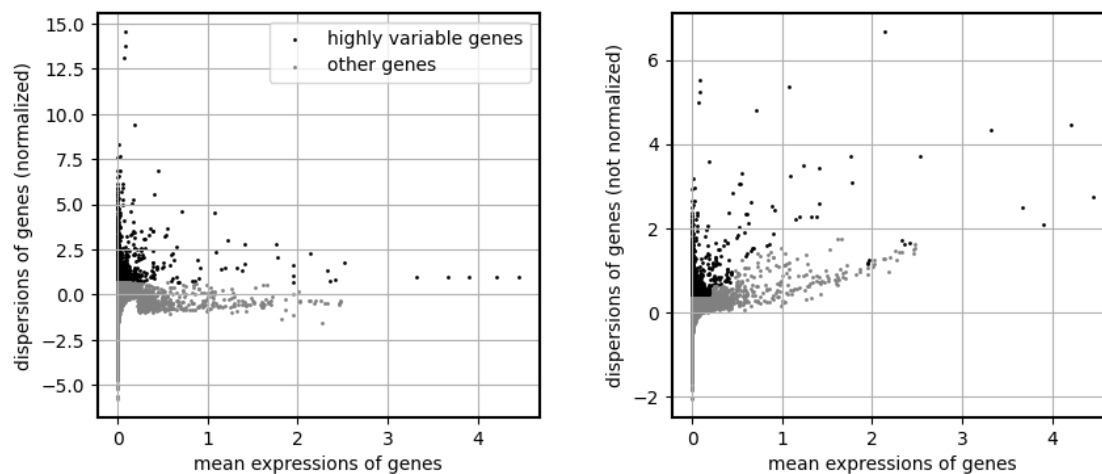
INFO:harmony:Iteration 3 of 10

2025-10-29 04:34:23,723 - harmony - INFO - Converged after 3 iterations

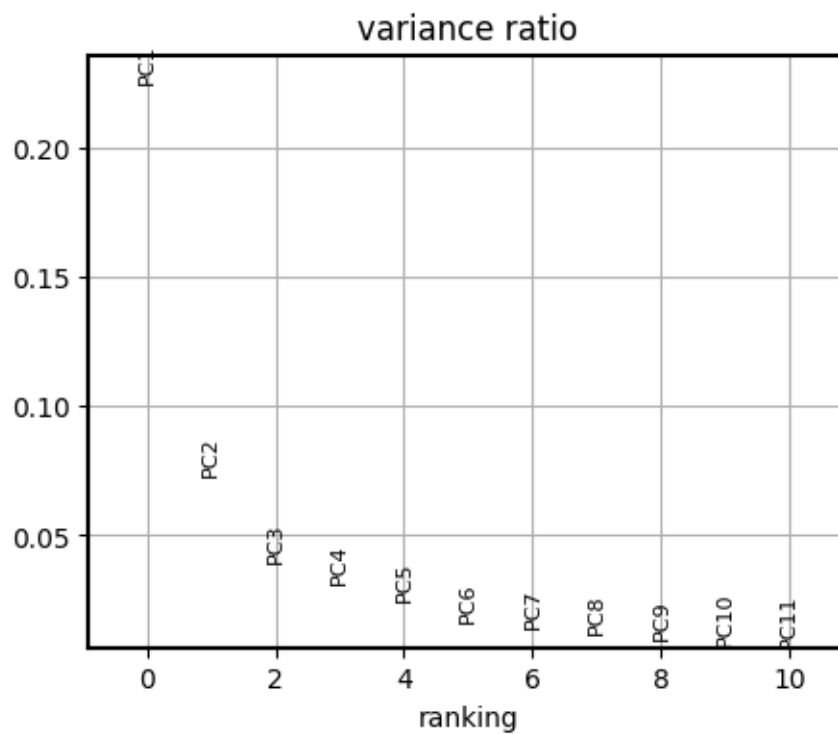
INFO:harmony:Converged after 3 iterations

```
[ ]: #Normalisation
ifn_adata.layers["counts"] = ifn_adata.X.copy()
sc.pp.normalize_total(ifn_adata)
sc.pp.log1p(ifn_adata)
```

```
[ ]: #Feature selection
sc.pp.highly_variable_genes(ifn_adata, n_top_genes=2500)
sc.pl.highly_variable_genes(ifn_adata)
```



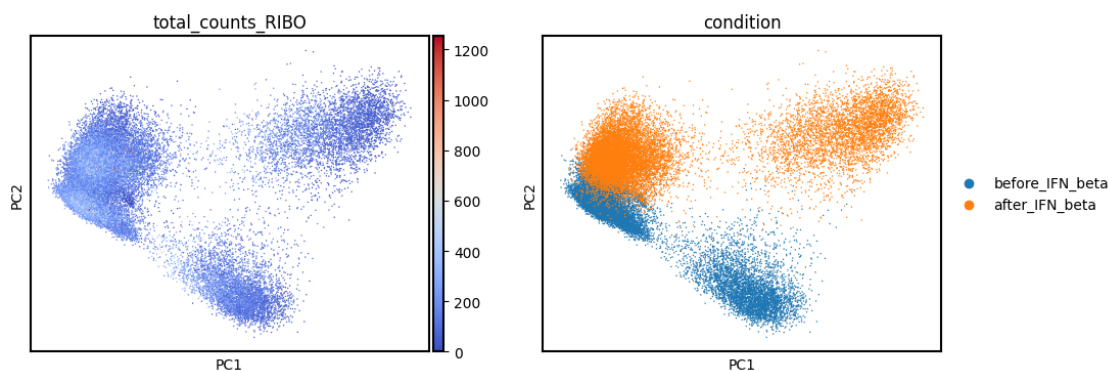
```
[ ]: #Dim Reduction
sc.tl.pca(ifn_adata)
sc.pl.pca_variance_ratio(ifn_adata, n_pcs=10, log=False)
```



```
[ ]: ifn_adata.var_names
```

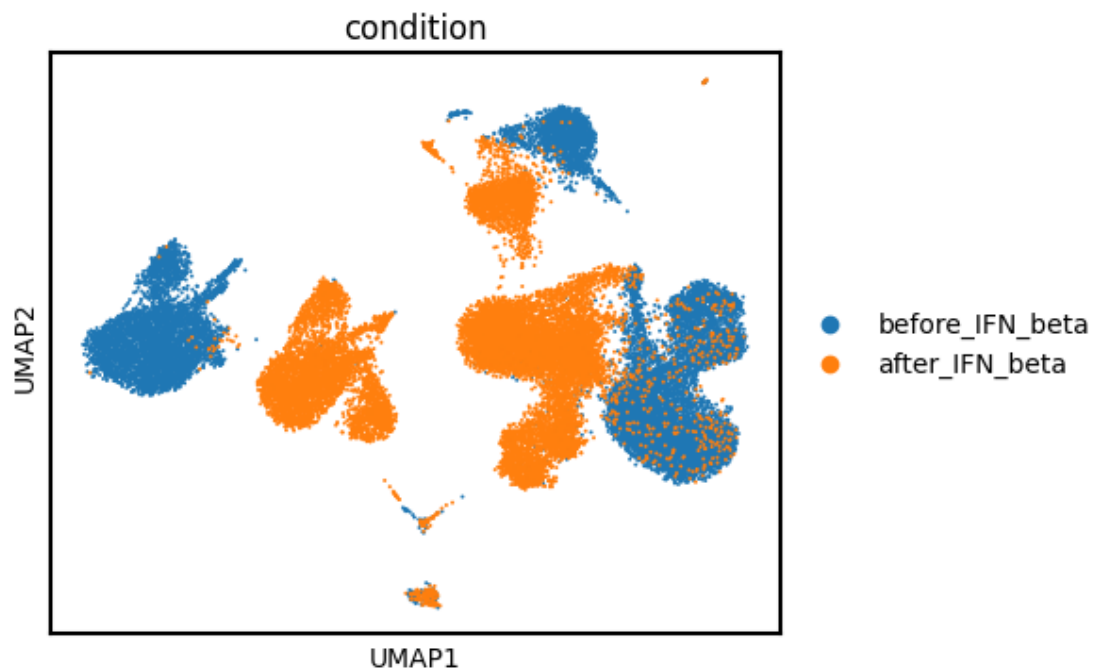
```
[ ]: Index(['MIR1302-10', 'FAM138A', 'OR4F5', 'RP11-34P13.7', 'RP11-34P13.8',
          'AL627309.1', 'RP11-34P13.14', 'RP11-34P13.9', 'AP006222.2',
          'RP4-669L17.10',
          ...,
          'MT-CO2', 'MT-ATP8', 'MT-ATP6', 'MT-CO3', 'MT-ND3', 'MT-ND4L', 'MT-ND4',
          'MT-ND5', 'MT-ND6', 'MT-CYB'],
          dtype='object', length=35635)
```

```
[ ]: sc.pl.pca(ifn_adata, cmap="coolwarm", color = ['total_counts_RIBO',
          ↪ 'condition'])
```



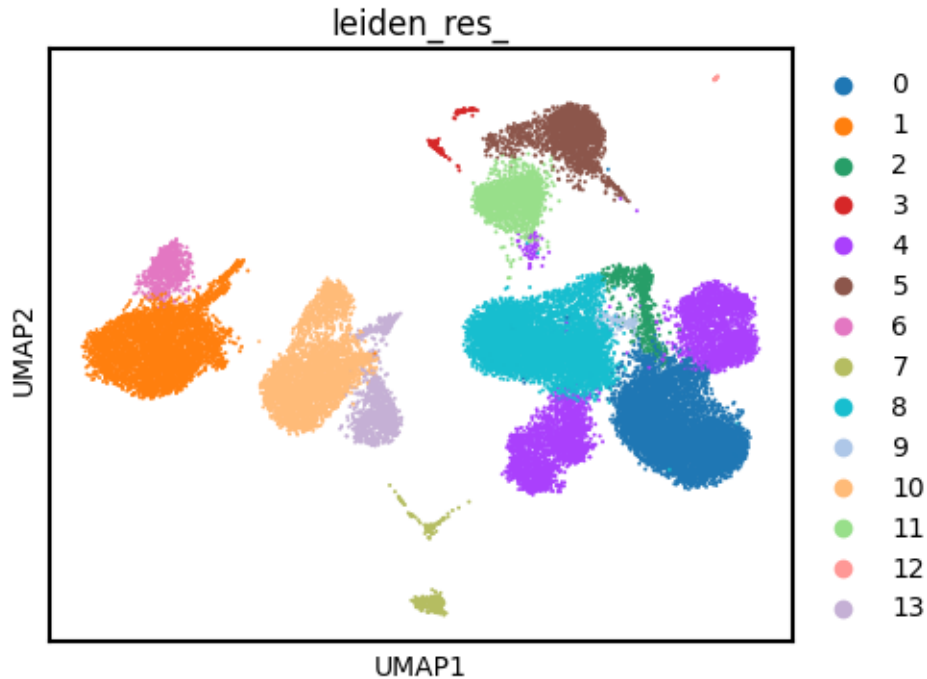
```
[ ]: sc.pp.neighbors(ifn_adata)
      sc.tl.umap(ifn_adata)
```

```
[ ]: sc.pl.umap(
      ifn_adata,
      size=8,
      color=["condition"],
      )
```



```
[ ]: sc.tl.leiden(ifn_adata, flavor="igraph", n_iterations=10, ↵  
    ↪key_added="leiden_res_", resolution=0.5 )
```

```
[ ]: sc.pl.umap(  
    ifn_adata,  
    color=["leiden_res_"],  
    size=8,  
)
```



```
[ ]: import decoupler as dc
```

```
[ ]: #!/wget wget -O result.txt 'http://www.ensembl.org/biomart/martservice?query=  

<?xml version="1.0" encoding="UTF-8"?><!DOCTYPE Query><Query  

<virtualSchemaName = "default" formatter = "CSV" header = "0" uniqueRows =  

<"0" count = "" datasetConfigVersion = "0.6" ><Dataset name =  

<"hsapiens_gene_ensembl" interface = "default" ><Attribute name =  

<"ensembl_gene_id" /><Attribute name = "external_gene_name" /></Dataset></  

<Query>'
```

```
[ ]: #import pandas as pd  

#ensembl_var = pd.read_csv('/content/result.txt', header = None)  

#ensembl_var.columns = ['ensembl_gene_id', 'gene_name']  

#ensembl_var.head(3)
```

```
[ ]: # Query Omnipath and get PanglaoDB  

markers = dc.op.resource(name="PanglaoDB", organism="human")  
  

# Remove duplicated entries  

markers = markers[~markers.duplicated(["cell_type", "genesymbol"])]  
  

#Format because dc only accepts cell_type and genesymbol
```

```

markers = markers.rename(columns={"cell_type": "source", "genesymbol": "target"})
markers = markers[["source", "target"]]

markers.head()

```

```
[ ]:          source  target
```

```

0    Hepatocytes    A1CF
1    Germ cells    A2APA5
2    Bergmann glia    A2M
3    Mast cells    A3FIN4
4    Mast cells    A4GALT

```

```
[ ]: ifn_adata.var_names
```

```
[ ]: Index(['MIR1302-10', 'FAM138A', 'OR4F5', 'RP11-34P13.7', 'RP11-34P13.8',
          'AL627309.1', 'RP11-34P13.14', 'RP11-34P13.9', 'AP006222.2',
          'RP4-669L17.10',
          ...,
          'MT-CO2', 'MT-ATP8', 'MT-ATP6', 'MT-CO3', 'MT-ND3', 'MT-ND4L', 'MT-ND4',
          'MT-ND5', 'MT-ND6', 'MT-CYB'],
          dtype='object', length=35635)
```

```
[ ]: dc.mt.ulm(data=ifn_adata,
              net=markers,
              tmin = 3)
```

```
[ ]: score = dc.pp.get_obsrm(ifn_adata, key="score_ulm")
```

```
[ ]: ifn_adata.obsrm["score_ulm"].head(1)
```

```
[ ]:
Acinar cells  Adipocyte progenitor cells  \
AAACATACAATGCC-1-before_IFN_beta      -0.80043      -0.60492

Adipocytes  Adrenergic neurons  \
AAACATACAATGCC-1-before_IFN_beta  -1.324387      -0.390399

Airway goblet cells  \
AAACATACAATGCC-1-before_IFN_beta      -0.653424

Airway smooth muscle cells  Alpha cells  \
AAACATACAATGCC-1-before_IFN_beta      -0.302386      0.41708

Alveolar macrophages  \
AAACATACAATGCC-1-before_IFN_beta      -0.819289

```

AAACATACAATGCC-1-before_IFN_beta	Anterior pituitary gland cells	Astrocytes \	0.63006	0.489759
AAACATACAATGCC-1-before_IFN_beta	... Taste receptor cells	Thymocytes \	0.162085	2.710003
AAACATACAATGCC-1-before_IFN_beta	Transient cells	Trichocytes \	-0.781119	-0.302386
AAACATACAATGCC-1-before_IFN_beta	Trigeminal neurons	Trophoblast cells \	-0.56289	-0.781119
AAACATACAATGCC-1-before_IFN_beta	Tuft cells	Undefined placental cells \	0.132304	-0.493861
AAACATACAATGCC-1-before_IFN_beta	Urothelial cells \		-0.493861	
AAACATACAATGCC-1-before_IFN_beta	Vascular smooth muscle cells		-0.349174	

[1 rows x 168 columns]

```
[ ]: ifn_adata.obsm["score_ulm"].columns
```

```
[ ]: Index(['Acinar cells', 'Adipocyte progenitor cells', 'Adipocytes',
          'Adrenergic neurons', 'Airway goblet cells',
          'Airway smooth muscle cells', 'Alpha cells', 'Alveolar macrophages',
          'Anterior pituitary gland cells', 'Astrocytes',
          ...
          'Taste receptor cells', 'Thymocytes', 'Transient cells', 'Trichocytes',
          'Trigeminal neurons', 'Trophoblast cells', 'Tuft cells',
          'Undefined placental cells', 'Urothelial cells',
          'Vascular smooth muscle cells'],
          dtype='object', length=168)
```

```
[ ]: #rank genes
ifn_gene_rank = dc.tl.rankby_group(score, groupby="leiden_res_",
    reference="rest", method="t-test_overestim_var")
ifn_gene_rank = ifn_gene_rank[ifn_gene_rank["stat"] > 0]
ifn_gene_rank.head(5)
```

[]:	group	reference	name	stat	meanchange	\
0	0	rest	T memory cells	42.618747	1.859396	
1	0	rest	Pancreatic progenitor cells	40.358809	1.426236	
2	0	rest	T follicular helper cells	40.199745	0.657304	

27	0	rest	T helper cells	38.039959	0.832729
29	0	rest	T cells naive	36.946212	2.236368

	pval	padj
0	0.000000e+00	0.000000e+00
1	0.000000e+00	0.000000e+00
2	0.000000e+00	0.000000e+00
27	5.802959e-299	3.481776e-298
29	3.216882e-283	1.801454e-282

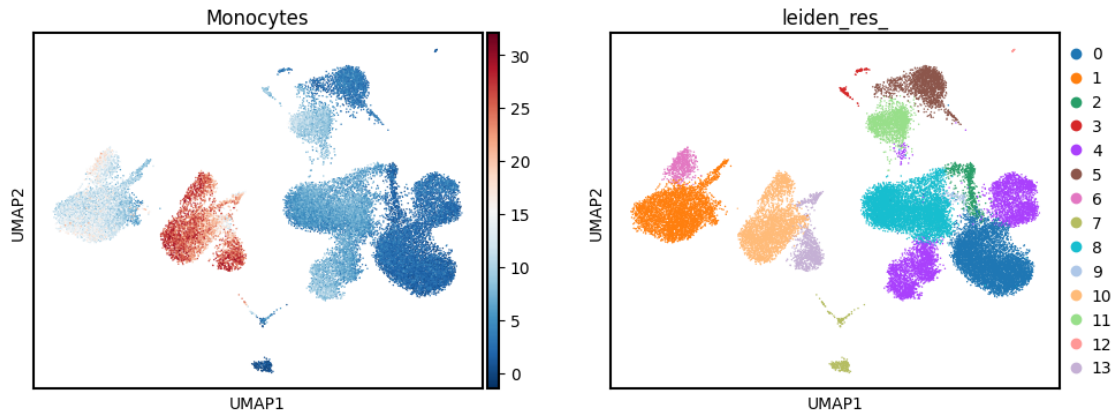
```
[ ]: top_names_per_group = ifn_gene_rank.groupby('group')['name'].apply(lambda x: x.
    ↪head(1))
display(top_names_per_group)
```

/tmp/ipython-input-4248453526.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
top_names_per_group = ifn_gene_rank.groupby('group')['name'].apply(lambda x:
x.head(1))
```

```
group
0      0      T memory cells
1     168    Microfold cells
10    1680      Monocytes
11    1848        B cells
12    2016  Pulmonary alveolar type II cells
13    2184      Monocytes
2     337      Melanocytes
3     504  Plasmacytoid dendritic cells
4     672        NK cells
5     840    B cells naive
6    1008    Dendritic cells
7    1176  Erythroid-like and erythroid precursor cells
8    1344      T cells naive
9    1512  Pulmonary alveolar type II cells
Name: name, dtype: category
Categories (168, object): ['Acinar cells', 'Adipocyte progenitor cells', ↪
    ↪'Adipocytes',
    ↪'Adrenergic neurons', ..., 'Tuft cells', 'Undefined',
    ↪'placental cells',
    ↪'Urothelial cells', 'Vascular smooth muscle cells']
```

```
[ ]: sc.pl.umap(score, color=["Monocytes", 'leiden_res_'], cmap="RdBu_r")
```



```
[ ]: n_ctypes = 1
      ctypes_dict = ifn_gene_rank.groupby("group").head(n_ctypes).
      ↪groupby("group")["name"].apply(lambda x: list(x)).to_dict()
      ctypes_dict
```

/tmp/ipython-input-4150618453.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
      ctypes_dict = ifn_gene_rank.groupby("group").head(n_ctypes).groupby("group")["
      name"].apply(lambda x: list(x)).to_dict()
```

```
[ ]: {'0': ['T memory cells'],
      '1': ['Microfold cells'],
      '10': ['Monocytes'],
      '11': ['B cells'],
      '12': ['Pulmonary alveolar type II cells'],
      '13': ['Monocytes'],
      '2': ['Melanocytes'],
      '3': ['Plasmacytoid dendritic cells'],
      '4': ['NK cells'],
      '5': ['B cells naive'],
      '6': ['Dendritic cells'],
      '7': ['Erythroid-like and erythroid precursor cells'],
      '8': ['T cells naive'],
      '9': ['Pulmonary alveolar type II cells']}
```

```
[ ]: dict_ann = ifn_gene_rank[ifn_gene_rank["stat"] > 0].groupby("group").head(1).
      ↪set_index("group")["name"].to_dict()
      dict_ann
```

/tmp/ipython-input-2298216777.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass

observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
dict_ann = ifn_gene_rank[ifn_gene_rank["stat"] >
0].groupby("group").head(1).set_index("group")["name"].to_dict()
```

```
[ ]: {'0': 'T memory cells',
      '1': 'Microfold cells',
      '2': 'Melanocytes',
      '3': 'Plasmacytoid dendritic cells',
      '4': 'NK cells',
      '5': 'B cells naive',
      '6': 'Dendritic cells',
      '7': 'Erythroid-like and erythroid precursor cells',
      '8': 'T cells naive',
      '9': 'Pulmonary alveolar type II cells',
      '10': 'Monocytes',
      '11': 'B cells',
      '12': 'Pulmonary alveolar type II cells',
      '13': 'Monocytes'}
```

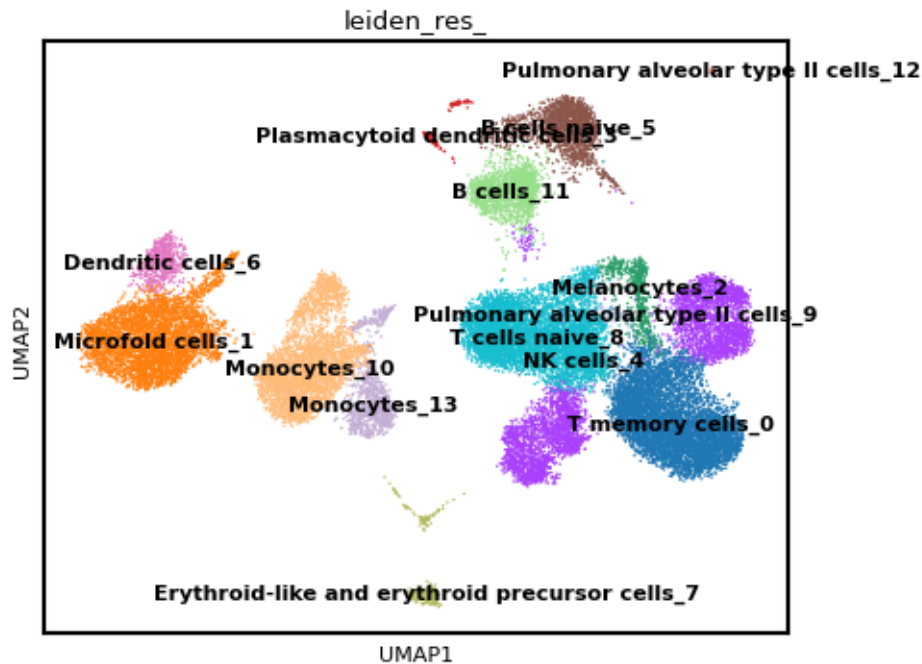
```
[ ]: dict_ann_unique = {k: v + '_' + str(k) for k, v in dict_ann.items()}
display(dict_ann_unique)
```

```
{'0': 'T memory cells_0',
 '1': 'Microfold cells_1',
 '2': 'Melanocytes_2',
 '3': 'Plasmacytoid dendritic cells_3',
 '4': 'NK cells_4',
 '5': 'B cells naive_5',
 '6': 'Dendritic cells_6',
 '7': 'Erythroid-like and erythroid precursor cells_7',
 '8': 'T cells naive_8',
 '9': 'Pulmonary alveolar type II cells_9',
 '10': 'Monocytes_10',
 '11': 'B cells_11',
 '12': 'Pulmonary alveolar type II cells_12',
 '13': 'Monocytes_13'}
```

```
[ ]: ifn_adata.obs["leiden_res_"] = ifn_adata.obs["leiden_res_"].cat.
      ↪rename_categories(dict_ann_unique)
```

```
[ ]: plt.rcParams["figure.figsize"] = (5,4) # Adjust figure size
      # Reduce the default font size for text elements in plots
      plt.rcParams['font.size'] = 8
      plt.rcParams['axes.labelsize'] = 8
      plt.rcParams['xtick.labelsize'] = 8
      plt.rcParams['ytick.labelsize'] = 8
```

```
# Now replot the UMAP
sc.pl.umap(
    adata=ifn_adata,
    color=[ "leiden_res_" ],
    ncols=2,
    legend_loc = 'on data',
    size=4,
)
```



1.0.6 Perturbation modelling

How it works: - **Input Data:** A single-cell expression matrix with cell-level metadata indicating the experimental condition (e.g., `condition = ['control', 'IFN_beta']`) and cell type annotations. - **Subsampling:** To avoid biases due to unequal cell numbers, Augur repeatedly subsamples a fixed number of cells per condition per cell type. - **Feature Selection:** It selects variable genes to focus on biologically relevant variation. - **Classification:** For each cell type, Augur trains a machine learning model—typically a random forest—to predict which condition each cell belongs to. - **Performance Scoring:** The mean AUC across subsampling iterations is computed. A high AUC indicates that the perturbation strongly shifts that cell type’s expression profile, making it easier to classify. A low AUC suggests minimal transcriptional change.

Here, we will use our dataset to know which cells are best targeted during stimulation IFN-stimulation

```
[ ]: ifn_adata.layers.keys()
```

```
[ ]: KeysView(Layers with keys: counts)
```

```
[ ]: raw_counts = ifn_adata.layers["counts"]
```

```
[ ]: raw_counts.shape
```

```
[ ]: (29065, 35635)
```

```
[ ]: raw_counts
```

```
[ ]: <Compressed Sparse Row sparse matrix of dtype 'float32'  
      with 17570845 stored elements and shape (29065, 35635)>
```

```
[ ]: ifn_adata.X = raw_counts
```

```
[ ]: ifn_adata.obs_keys()
```

```
/tmp/ipython-input-969156741.py:1: FutureWarning: Use obs (e.g. `k` in adata.obs`  
or `str(adata.obs.columns.tolist())` instead of AnnData.obs_keys`  
AnnData.obs_keys is deprecated and will be removed in the future.  
    ifn_adata.obs_keys()
```

```
[ ]: ['condition',  
      'n_genes_by_counts',  
      'log1p_n_genes_by_counts',  
      'total_counts',  
      'log1p_total_counts',  
      'pct_counts_in_top_50_genes',  
      'pct_counts_in_top_100_genes',  
      'pct_counts_in_top_200_genes',  
      'pct_counts_in_top_500_genes',  
      'total_counts_MT',  
      'log1p_total_counts_MT',  
      'pct_counts_MT',  
      'total_counts_RIBO',  
      'log1p_total_counts_RIBO',  
      'pct_counts_RIBO',  
      'total_counts_HB',  
      'log1p_total_counts_HB',  
      'pct_counts_HB',  
      'leiden_res_']
```

```
[ ]: ifn_adata.obs["condition"]
```

```
[ ]: AAACATACAATGCC-1-before_IFN_beta    before_IFN_beta
AAACATACATTTCC-1-before_IFN_beta        before_IFN_beta
AAACATACCAGAAA-1-before_IFN_beta        before_IFN_beta
AAACATACCAGCTA-1-before_IFN_beta        before_IFN_beta
AAACATACCATGCA-1-before_IFN_beta        before_IFN_beta
...
TTTGCATGCTAAGC-1-after_IFN_beta         after_IFN_beta
TTTGCATGGGACGA-1-after_IFN_beta         after_IFN_beta
TTTGCATGGTGAGG-1-after_IFN_beta         after_IFN_beta
TTTGCATGGTTTGG-1-after_IFN_beta         after_IFN_beta
TTTGCATGTCTTAC-1-after_IFN_beta         after_IFN_beta
Name: condition, Length: 29065, dtype: category
Categories (2, object): ['before_IFN_beta', 'after_IFN_beta']
```

```
[ ]: ifn_adata.obs.leiden_res_.value_counts()
```

```
[ ]: leiden_res_
T memory cells_0                6089
T cells naive_8                  5834
NK cells_4                       4260
Microfold cells_1                3895
Monocytes_10                    3209
B cells naive_5                  1708
B cells_11                      1477
Monocytes_13                     962
Dendritic cells_6                570
Melanocytes_2                    559
Erythroid-like and erythroid precursor cells_7  283
Plasmacytoid dendritic cells_3    118
Pulmonary alveolar type II cells_9   82
Pulmonary alveolar type II cells_12  19
Name: count, dtype: int64
```

```
[ ]: ag_rfc = pt.tl.Augur("random_forest_classifier")
```

```
[ ]: loaded_data = ag_rfc.load(ifn_adata, label_col="condition",
    ↪ cell_type_col="leiden_res_")
loaded_data
```

```
[ ]: AnnData object with n_obs × n_vars = 29065 × 35635
      obs: 'label', 'n_genes_by_counts', 'log1p_n_genes_by_counts',
'total_counts', 'log1p_total_counts', 'pct_counts_in_top_50_genes',
'pct_counts_in_top_100_genes', 'pct_counts_in_top_200_genes',
'pct_counts_in_top_500_genes', 'total_counts_MT', 'log1p_total_counts_MT',
'pct_counts_MT', 'total_counts_RIBO', 'log1p_total_counts_RIBO',
'pct_counts_RIBO', 'total_counts_HB', 'log1p_total_counts_HB', 'pct_counts_HB',
'cell_type', 'y_'
```

```

var: 'MT', 'RIBO', 'HB', 'n_cells_by_counts', 'mean_counts',
'log1p_mean_counts', 'pct_dropout_by_counts', 'total_counts',
'log1p_total_counts', 'highly_variable', 'means', 'dispersions',
'dispersions_norm'
layers: 'counts'

```

```

[ ]: v_adata, v_results = ag_rfc.predict(
    loaded_data, subsample_size=50, n_threads=4, select_variance_features=True,
    span=1
)

v_results["summary_metrics"]

```

! Set smaller span value in the case of a `segmentation fault` error.
! Set larger span in case of svddc or other near singularities error.

Output()

! Skipping Microfold cells_1 cell type - the number of samples for at least one class type is less than subsample size 50.
! Skipping Plasmacytoid dendritic cells_3 cell type - the number of samples for at least one class type is less than subsample size 50.
! Skipping Dendritic cells_6 cell type - the number of samples for at least one class type is less than subsample size 50.
! Skipping Pulmonary alveolar type II cells_9 cell type - the number of samples for at least one class type is less than subsample size 50.
! Skipping Monocytes_10 cell type - the number of samples for at least one class type is less than subsample size 50.
! Skipping B cells_11 cell type - the number of samples for at least one class type is less than subsample size 50.
! Skipping Pulmonary alveolar type II cells_12 cell type - 19 samples is less than min_cells 50.
! Skipping Monocytes_13 cell type - the number of samples for at least one class type is less than subsample size 50.

```

[ ]:
      T memory cells_0  Melanocytes_2  NK cells_4  \
mean_augur_score      0.619937      0.647404      0.701807
mean_auc              0.619937      0.647404      0.701807
mean_accuracy         0.575152      0.594575      0.636762
mean_precision        0.627809      0.637931      0.663549
mean_f1               0.466391      0.532513      0.610382
mean_recall           0.393578      0.484093      0.599069

      B cells naive_5  \
mean_augur_score      0.839610
mean_auc              0.839610
mean_accuracy         0.758241

```

```
mean_precision      0.720572
mean_f1             0.781923
mean_recall         0.865441
```

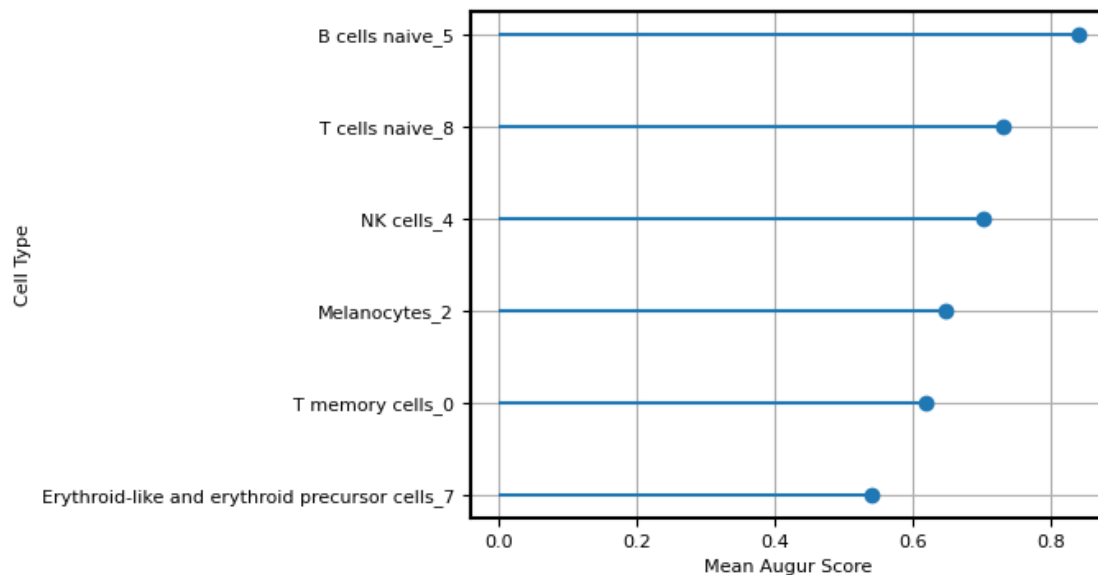
```

Erythroid-like and erythroid precursor cells_7 \
mean_augur_score      0.539469
mean_auc              0.539469
mean_accuracy         0.534528
mean_precision        0.538655
mean_f1               0.530117
mean_recall           0.552843
```

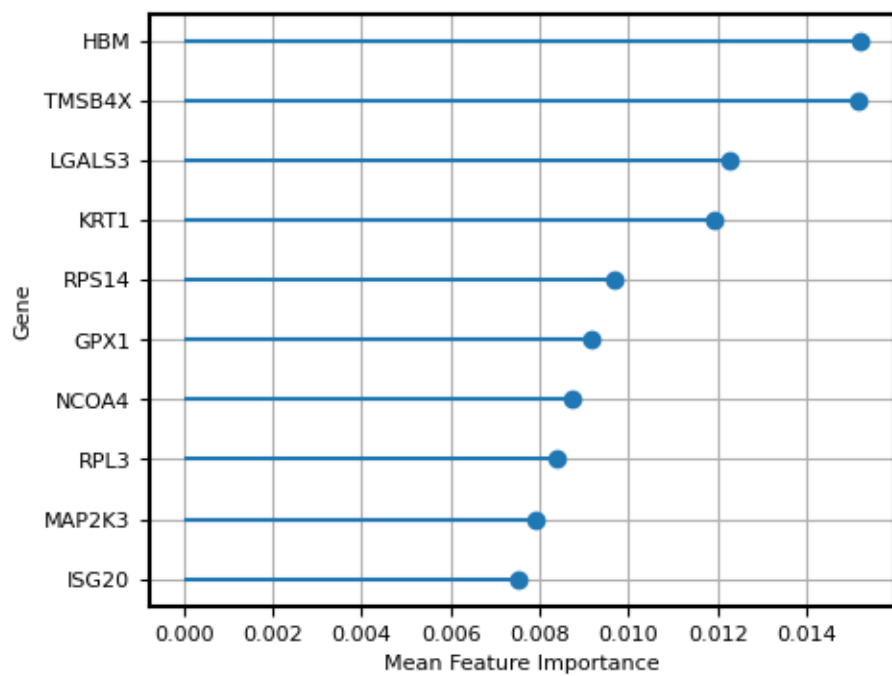
```

T cells naive_8
mean_augur_score      0.730509
mean_auc              0.730509
mean_accuracy         0.659257
mean_precision        0.670737
mean_f1               0.649675
mean_recall           0.652843
```

```
[ ]: lollipop = ag_rfc.plot_lollipop(v_results)
```



```
[ ]: important_features = ag_rfc.plot_important_features(v_results)
```

[]: